

Ultra-Low-Power PMU

IEEE • SI-CLASH

CMOS 65nm technology

POR

1. Executive summary

This week we implemented and simulated a resistor-divider + differential-comparator POR with resistive positive feedback (hysteresis), sized with the gm/Id methodology on a 65 nm NMOS lookup table (TSMC 65 nm, crn65gplus 2.5 V devices, Spectre, typical corner, 27 °C). The circuit produces a clean, hysteretic POR transition close to the required battery window. The latest transient simulation (7 October) gives:

- **Rising threshold (Vth_high): 2.481 V** (transient marker: 2.4809 V at 1.4886 ms) against a limit of ≤ 2.5 V — met, 19 mV margin.
- **Falling threshold (Vth_low): 2.288 V** (transient marker: 2.2876 V at 8.6274 ms) against a limit of ≥ 2.3 V — **not met**, 12 mV below the limit.
- **Hysteresis: ≈ 193 mV** (2.4809 − 2.2876 V) against a limit of ≤ 200 mV — met, but only 7 mV margin.
- **Current consumption (R.033): 0.55 µA at VBAT = 2.3 V and 0.66 µA at 3.2 V** (DC sweep of the VBAT source current) against a 1 µA budget — met.

Because the hysteresis (193 mV) almost fills the 200 mV threshold window (2.3–2.5 V), the two thresholds can only be inside the window if the rising threshold sits between about 2.493 V and 2.5 V. Reducing the hysteresis to about 100–150 mV and re-centring the thresholds (≈ 2.45 V / 2.33 V) is therefore the first design action.

VREF source. The reference source for the comparator has not been determined yet. In the simulations VREF is an ideal 0.974 V DC source; in the final design it will most likely be provided by the bandgap reference, but this is not confirmed and must be agreed at PMU top level.

Not yet demonstrated: PVT / Monte-Carlo behaviour and the power-down current (R.004 ≤ 1 nA; no enable switch yet). Our main technical finding (Section 6.4) is that the comparator does not currently operate as a balanced differential pair, so the threshold is likely set by the M0 gate-source voltage rather than by VREF. This must be fixed before PVT/MC sign-off and drives the next steps in Section 7.

Status at a glance

Req.	Parameter	Spec	Result (TT, 27 °C)	Status
R.034	Vth_high (POR → 1)	≤ 2.5 V	2.481 V (transient)	Pass (19 mV margin)
R.035	Vth_low (POR → 0)	≥ 2.3 V	2.288 V (transient)	Fail (−12 mV)
R.036	Hysteresis	≤ 200 mV	≈ 193 mV	Pass (7 mV margin)
R.033	Current consumption	Budgeted ≤ 1 µA	0.55 µA @ 2.3 V; 0.66 µA @ 3.2 V (simulated)	Pass
R.004	Power-down current	≤ 1 nA	Not implemented (no enable)	Open
—	PVT / Monte Carlo	Checkpoint 5	Not started	Open
—	VREF source	—	Ideal 0.974 V source; likely bandgap, not yet decided	Open

2. Requirements and budget derivation

The POR must monitor the battery (2.3–3.2 V range, R.001) and control PMU start-up/shutdown with the following thresholds (R.033–R.036):

Req.	Symbol	Limit	Meaning
R.034	Vth_high	≤ 2.5 V	Above this VBAT the POR output is '1'
R.035	Vth_low	≥ 2.3 V	Below this VBAT the POR output is '0'
R.036	Vhyst	≤ 200 mV	Vth_high − Vth_low
R.033	Icon	budget	Use R.002 and R.009 to budget

2.1 Current budget for the POR (R.002 → R.033)

$$T = 90\ \mathrm{d} \times 24\ \mathrm{h/d} = 2160\ \mathrm{h}$$

$$I_{avg,max} = 150\ \mathrm{mAh}\ /\ 2160\ \mathrm{h} = 69.4\ \mu\mathrm{A}$$

$$I_{rem} = I_{avg,max} - I_{load,max} = 69.4 - 50 = 19.4\ \mu\mathrm{A}$$

$$I_{POR} \leq 1\ \mu\mathrm{A} \approx 1.4\ \%\ I_{avg,max}$$

The threshold window is only 200 mV wide (2.3–2.5 V, i.e. ±4 % around 2.4 V), so accuracy, not power, is the dominant design tension for this block.

3. Architecture and polarity convention

The battery is scaled by a resistive divider (R1/R0) to node vmon. A degenerated NMOS differential pair (M0 on vmon, M1 on VREF) with resistive loads (R4, R5, 150 MΩ) forms the first stage; a tail current mirror (M3 diode-connected, M2 mirror, I5 = 100 nA) sets the bias. M4 with load R2 is the second-stage inverter that produces the output. R3 feeds the output back to vmon, so the divider ratio changes with the output state, which creates the hysteresis.

Reference source: the origin of VREF is not determined yet. The simulations use an ideal 0.974 V source (V1); the reference will most likely come from the bandgap, which still has to be confirmed at PMU level.

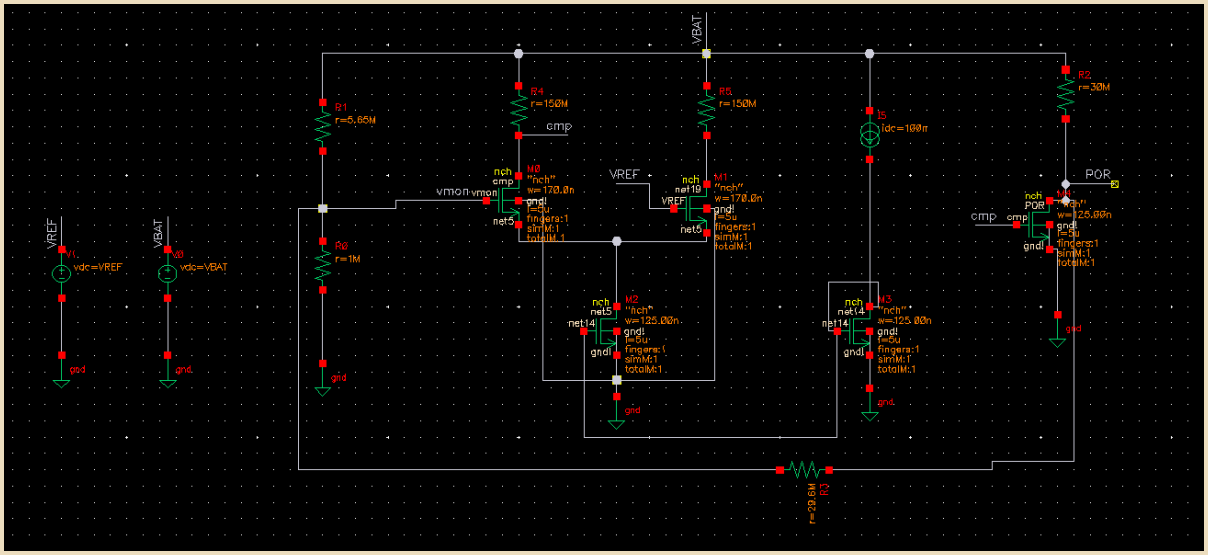


Figure 1 — POR2 schematic (as simulated)

3.1 Polarity convention and feedback direction

The comparator-side (internal) decision signal must follow this convention, because the feedback NMOS (M_FB) needs it to be ON below the trip point:

Condition	Internal decision node (POR_INT / cmp)	Feedback NMOS M_FB
VBAT < threshold	1	ON
VBAT > threshold	0	OFF

This gives the correct hysteresis direction: while VBAT is rising and still below the trip point, M_FB is ON and shunts the divider so vmon is lower, so a **higher** VBAT is needed to trip (Vth_high). Once tripped, M_FB turns OFF, vmon is relatively higher, and the battery must fall further to release (Vth_low < Vth_high). In the simulated netlist the cmp node (drain of M0/R4) already has exactly this polarity (cmp = 1 below threshold, 0 above).

> **Polarity note:** the external POR pin required by the spec (R.034/R.035) is the opposite sense — POR = '1' above Vth_high and '0' below Vth_low. In the current netlist that pin is the M4/R2 inverter output, i.e. POR = NOT(cmp). Our simulated plots therefore show POR = 0 below and POR = 1 above the threshold, as the spec requires. In the current version the hysteresis is realised by the resistor R3 driven from the POR pin (a POR = 1 state raises the divider ratio, the same effect as switching M_FB OFF). If the resistive feedback is replaced by an explicit NMOS M_FB driven by POR_INT, the two implementations are equivalent. Please confirm which polarity the top-level PMU integration needs at the output pin.

4. gm/Id sizing methodology and calculations

Devices were sized from a TSMC 65 nm NMOS gm/Id look-up table swept over VGS (0.2–1.0 V) for L = 0.06 µm … 5 µm. The methodology replaces square-law hand analysis (invalid in the weak/moderate inversion region where a nano-power POR operates) by reading gm/Id, VGS and current density directly from the LUT.

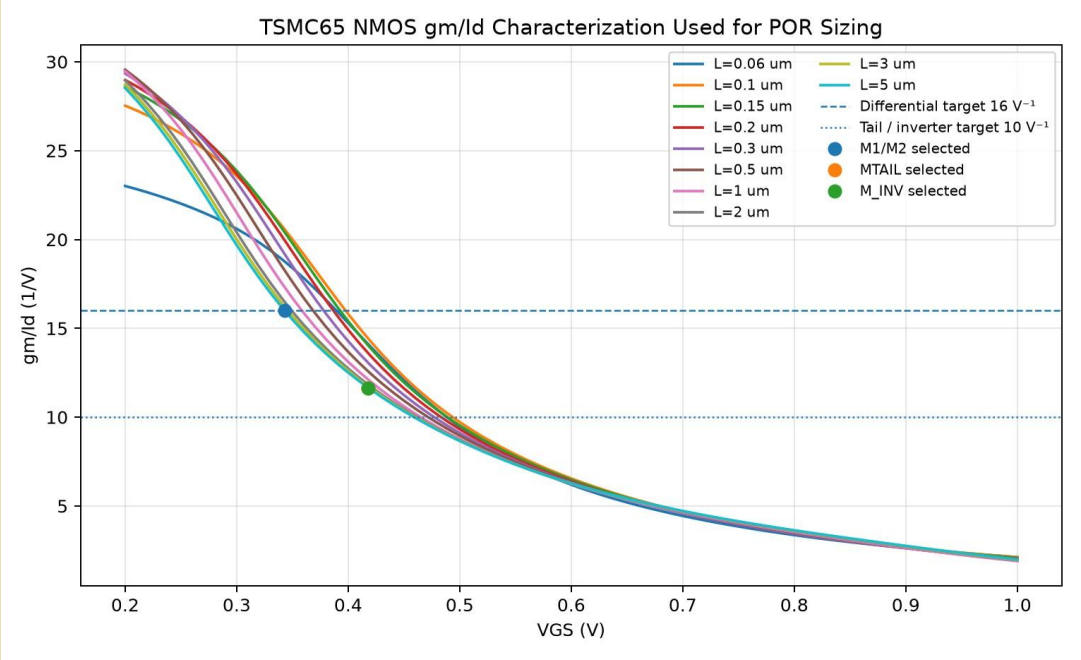


Figure 2 — TSMC65 NMOS gm/Id characterization used for POR sizing (selected operating points marked)

4.1 Step-by-step procedure

1. **Fix currents.** I_{tail} = 100 nA (mirror I5 → M3 → M2). Balanced operation gives I_D(M0) = I_D(M1) = 50 nA. The mirror and the inverter reference branch carry 100 nA.

2. **Pick gm/Id per role.** Differential pair: 16 V^{-1} (weak/moderate inversion, maximum gain per nA and low VGS). Tail mirror and inverter: 10 V^{-1} target (more overdrive, better mirror matching, less noise/leakage sensitivity).
3. **Pick L = 5 µm for all devices.** Long channel gives high ro (gain from the resistive stage stays ≈ R_{load}), better matching ($\sigma VT \propto 1/\sqrt{(WL)}$), and the curves for L ≥ 1 µm collapse onto one gm/Id–VGS relation, which makes the read-out insensitive to L.
4. **Read VGS from the LUT** at the chosen gm/Id and L = 5 µm.
5. **Compute gm, headroom and W/L** from current density (I_D/W) read from the LUT: $W = I_D / (I_D/W)|_{LUT}$. Figure 3 shows the I_D/W curves for each channel length, the legal I_D/W bands for 50 nA and 100 nA with W = 0.12–20 µm, and the chosen operating points.
6. **Check gain, offset and current** analytically (Sections 4.3–4.5).

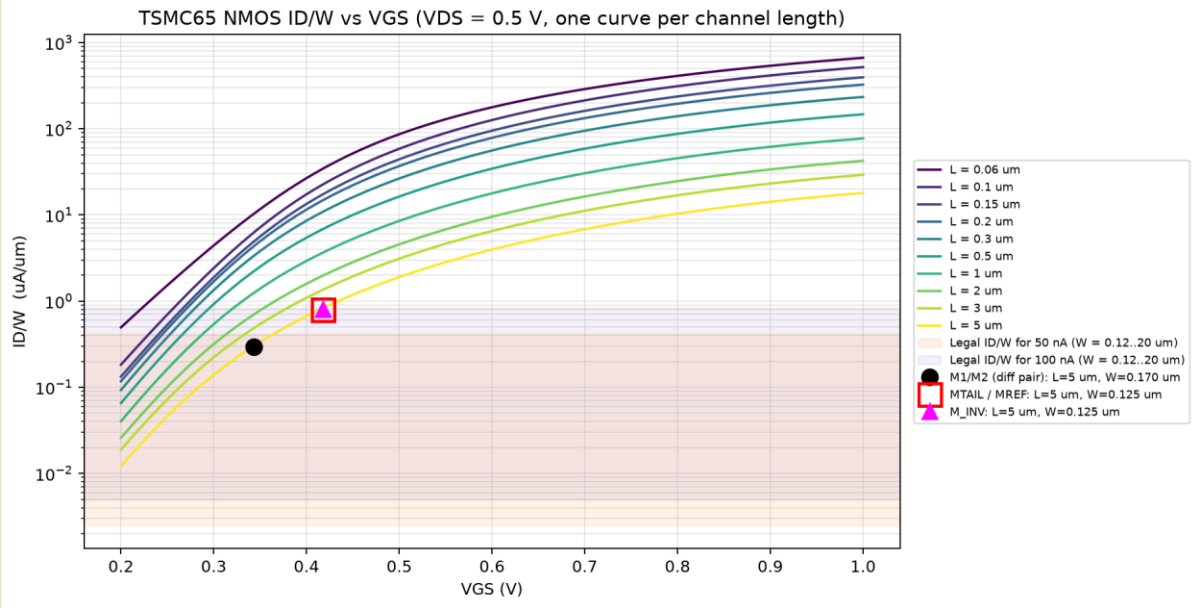


Figure 3 — TSMC65 NMOS ID/W vs VGS (VDS = 0.5 V, one curve per channel length) with the selected operating points

4.2 Operating points and device sizes

Role (netlist name)	Chosen gm/Id (V^{-1})	VGS (V)	I_D (nA)	gm = (gm/Id)·I_D	W/L (drawn)	Implied I_D per W/L
Diff. pair M0 (vmon) / M1 (VREF)	16	0.343	50	0.80 µS	170 nm / 5 µm = 0.034	1.47 µA
Tail M2 (mirror) / M3 (diode)	≈ 11.6 (target 10)	0.418	100	1.16 µS	125 nm / 5 µm = 0.025	4.0 µA
Inverter M4	≈ 11.6 (target 10)	0.418	≈ 50–100	≈ 0.6–1.2 µS	125 nm / 5 µm = 0.025	4.0 µA

Note: the LUT plots name the differential pair "M1/M2"; in the schematic these are M0/M1. The tail and inverter share one operating point; the tail marker is hidden underneath it.

$$g_{m,pair} = (g_m/I_D) \cdot I_D = 16\ V^{-1} \times 50\ nA = 0.80\ \mu S$$

$$g_{m,tail} = 11.6\ V^{-1} \times 100\ nA = 1.16\ \mu S$$

$$V_{dsat,pair} \approx 2\ /\ (g_m/I_D) = 2\ /\ 16\ V^{-1} = 0.125\ V$$

$$V_{dsat,tail} \approx 2\ /\ 11.6\ V^{-1} = 0.17\ V$$

$$V_{net5} = V_{REF} - V_{GS} = 0.974 - 0.343 = 0.631\ V$$

$$V_{DS,tail} = 0.631\ V > V_{dsat,tail} = 0.17\ V$$

The tail therefore stays in saturation with about 0.46 V of headroom (balanced-pair case).

4.3 Sanity check of current density (EKV model)

With n ≈ 1.3 and U_T = 25.9 mV, $g_m/I_D = 1/(n \cdot U_T) / (0.5 + \sqrt{(0.25 + IC)})$ gives the inversion coefficient IC, and the specific current $I_S = (I_D/(W/L))/IC$ should be the same for both devices:

$$IC = ((1/(n \cdot U_T)) / (g_m/I_D) - 0.5)^2 - 0.25$$

$$IC_{pair} = (29.7/16 - 0.5)^2 - 0.25 = 1.6, \quad IC_{tail} = (29.7/11.6 - 0.5)^2 - 0.25 = 4.0$$

$$I_{S,pair} = 1.47\ \mu A / 1.6 = 0.92\ \mu A, \quad I_{S,tail} = 4.0\ \mu A / 4.0 = 1.0\ \mu A$$

Both devices give a specific current of about 1 µA, so the drawn W/L values are consistent with the LUT.

4.4 Gain of the comparator chain (balanced-pair assumption)

$$r_o \approx 1/(\lambda \cdot I_D) = 1 / (0.05\ V^{-1} \times 50\ nA) = 400\ M\Omega$$

$$R4 \parallel r_o = (150 \times 400)/(150 + 400) = 109\ M\Omega$$

$$A_{v1} = g_{m,pair}\ (R4 \parallel r_o) = 0.80\ \mu S \times 109\ M\Omega = 87\ V/V\ (38.8\ dB)$$

$$A_{v2} = g_{m4}\ (R2 \parallel r_o) = 0.58\ \mu S \times 27.9\ M\Omega = 16\ V/V$$

$$A = A_{v1} \cdot A_{v2} \approx 87 \times 16 \approx 1400\ V/V\ (63\ dB)$$

$$\Delta v_{mon} = 1.4\ V / 1400 = 1\ mV, \quad \Delta V_{BAT} = \Delta v_{mon}/\beta_L \approx 7\ mV$$

This agrees with the near-vertical jump seen in the DC sweep (Figures 4–5).

4.5 Offset from mismatch (expected for a balanced pair)

$$\sigma(\Delta V_T) = A_{VT}/\sqrt{(WL)} = 3.5\ mV \cdot \mu m / \sqrt{(0.85\ \mu m^2)} = 3.8\ mV$$

$$\sigma(V_{BAT}) = \sigma(\Delta V_T)/\beta_L = 3.8\ mV / 0.146 = 26\ mV, \quad 3\sigma = 78\ mV$$

Three-sigma offset alone would consume ~40 % of the 200 mV threshold window, which is the reason a trim (TRIM<3:0> on the divider) and a Monte-Carlo run are planned.

5. Threshold and hysteresis calculation

Divider values: R1 = 5.65 MΩ, R0 = 1 MΩ, R3 = 29.6 MΩ (feedback), R2 = 30 MΩ (pull-up). Two divider states exist depending on the POR output level:

5.1 POR = 0 (below threshold): R3 in parallel with R0

$$R0 \parallel R3 = (R0 \cdot R3)/(R0 + R3) = (1 \times 29.6)/(1 + 29.6)\ M\Omega = 0.9673\ M\Omega$$

$$\beta_L = (R0 \parallel R3)/(R1 + R0 \parallel R3) = 0.9673/(5.65 + 0.9673) = 0.1462$$

$$V_{BAT,rise} = V_{trip} / \beta_L$$

5.2 POR = 1 (above threshold): R3 returns to the high output

Solving the two node equations (M4 off) at vmon = Vtrip:

$$P = (R3 \cdot V_{BAT} + R2 \cdot V_{trip})/(R2 + R3) = (29.6 \cdot V_{BAT} + 30 \cdot V_{trip})/59.6$$

Kirchhoff's current law at node vmon gives:

$$(V_{BAT} - V_{trip})/R1 = V_{trip}/R0 + (V_{trip} - P)/R3$$

Substituting P and simplifying:

$$0.1938\,(V_{BAT} - V_{trip}) = V_{trip} \;\Rightarrow\; V_{BAT,fall} = 6.161 \cdot V_{trip},\quad \beta_H = 0.1623$$

$$V_{BAT,rise} \,/\, V_{BAT,fall} = \beta_H \,/\, \beta_L = 0.1623 \,/\, 0.1462 = 1.110$$

$$V^{ideal}_{BAT,fall} = 2.50\ V \,/\, 1.110 = 2.25\ V,\quad V^{ideal}_{hyst} = 2.50 - 2.25 \approx 0.249\ V$$

The ideal-comparator hysteresis (≈ 249 mV) would exceed the 200 mV limit. The latest simulation (transient, Figure 6) gives:

$$2.4809\ V \,/\, 2.2876\ V = 1.0845,\quad V_{hyst} = 2.4809 - 2.2876 \approx 193\ mV$$

The simulated hysteresis is smaller than the ideal-comparator prediction because the comparator trip point is not a fixed vmon (the second-stage inverter, POR high level and pair operating region all shift with the output state). Section 6.4 explains the main reason. The simulated value (193 mV) meets the ≤ 200 mV limit but leaves only 7 mV of margin.

5.3 Battery current of the POR

Hand estimate:

$$I_{div} = V_{BAT} \,/\, (R1 + R0 \parallel R3) = 3.0\ V \,/\, 6.62\ M\Omega = 0.45\ \mu A$$

$$I_{div}(3.2\ V) = 3.2\ V \,/\, 6.62\ M\Omega = 0.48\ \mu A$$

$$I_{POR} = I_{div} + I_{tail} + I_{R2,R4,R5} \approx 0.45 + 0.10 + (0.03\text{–}0.1) \approx 0.65\text{–}0.75\ \mu A$$

The mirror branch can add up to a further 0.10 µA if it conducts fully. The simulated result in Section 6.5 (0.55–0.66 µA over VBAT = 2.3–3.2 V) confirms that the total is below the 1 µA allocation.

6. Simulation results versus specification

6.1 DC sweeps (typical corner)

The DC sweeps below (run on 29 September, earlier netlist iteration) show the output when VBAT is swept through the threshold in each direction. The output stays at 0 V until the threshold and then jumps abruptly (≈ 1.4 V just above the rising threshold) and continues rising with VBAT to ≈ 2.35 V at 5 V. These sweeps predate the 7 October transient run and should be re-run on the current netlist so that DC and transient thresholds can be compared on the same design.

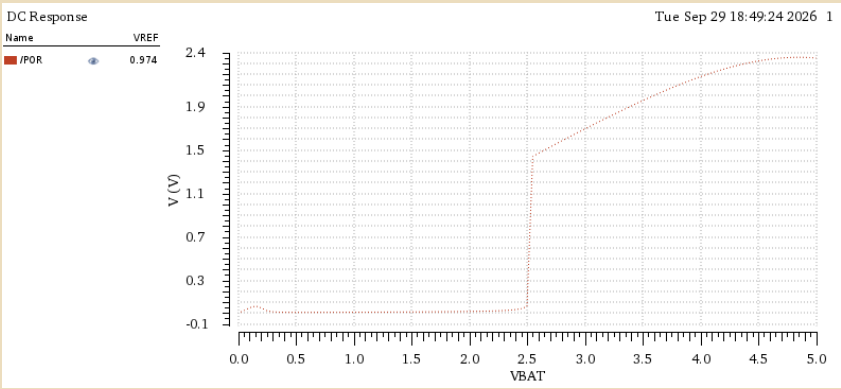


Figure 4 — POR vs. rising VBAT: transition at ≈ 2.5 V

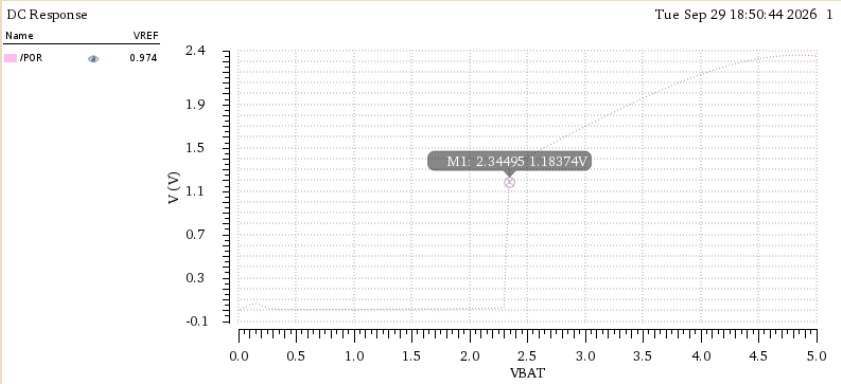


Figure 5 — POR vs. falling VBAT: transition at 2.345 V (cursor M1)

6.2 Transient hysteresis check

A 0 → 5 V ramp (3 ms), 4 ms plateau and falling ramp shows the POR asserting at 1.4886 ms (cursor M1, VBAT = 2.4809 V) and releasing at 8.6274 ms (cursor M2, VBAT = 2.2876 V). This gives Vth_high = 2.4809 V and Vth_low = 2.2876 V, i.e. a hysteresis of 2.4809 − 2.2876 = 193.3 mV.

The rising threshold meets the ≤ 2.5 V limit (19 mV margin), and the hysteresis meets the ≤ 200 mV limit. The falling threshold, however, lies 12.4 mV below the ≥ 2.3 V limit. The DC sweep of 29 September gave a higher falling threshold (2.345 V); part of the difference may come from the finite ramp rate (≈ 1.67 V/ms on the falling edge) and part from netlist changes, which must be separated by re-running the DC sweep on the current netlist.

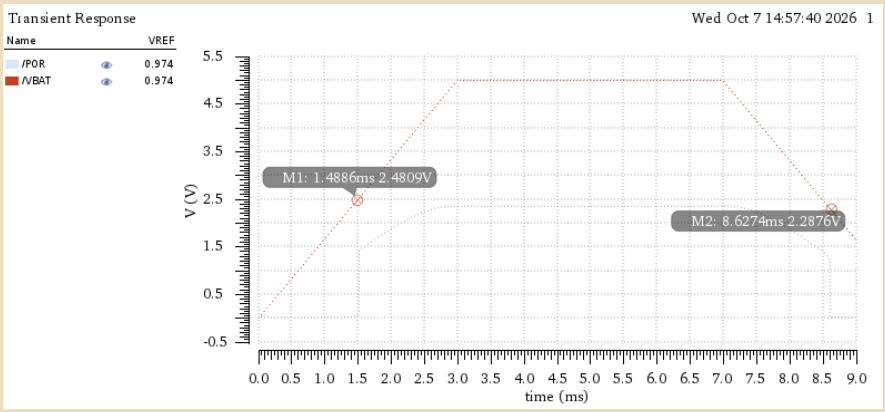


Figure 6 — Transient hysteresis test (VBAT ramp up / down, POR output); M1: 1.4886 ms, 2.4809 V; M2: 8.6274 ms, 2.2876 V

6.3 Comparison table

Req.	Parameter	Spec	Result	Comment
R.034	Vth_high	≤ 2.5 V	2.481 V (transient)	Meets limit with 19 mV margin; PVT will likely push it above 2.5 V without trim / re-centring to ≈ 2.45 V
R.035	Vth_low	≥ 2.3 V	2.288 V (transient)	Fails by 12 mV; DC value (29 Sep netlist) was 2.345 V, so DC re-run required; re-centre to ≈ 2.33 V
R.036	Vhyst	≤ 200 mV	≈ 193 mV	Meets, but only 7 mV margin and it leaves no room for both thresholds inside 2.3–2.5 V; target ≈ 100–150 mV

Req.	Parameter	Spec	Result	Comment
R.033	Icon	≤ 1 µA (budget)	0.55 µA @ 2.3 V; 0.66 µA @ 3.2 V	Simulated (Figure 7); slightly below the hand estimate of 0.65–0.75 µA; ≈ 0.95 % of Iavg,max at 3.2 V
R.004	Ipd	≤ 1 nA	n/a	No enable / power-down path yet; the POR draws ≈ 0.55–0.66 µA at all times
—	POR high level	Digital-usable	≈ 1.4 V at VBAT = 2.55 V	Only ~55 % of VBAT; needs an output buffer
—	VREF source	—	Ideal 0.974 V source	Source not yet determined; most likely the bandgap

6.4 Key finding: the comparator is current-starved

> **Design risk found this week.** If VREF = 0.974 V were the comparison reference, the trip point would be VBAT = VREF / β_L = 0.974 / 0.1462 = 6.66 V, yet the simulation trips at ≈ 2.5 V (vmon ≈ 0.365 V). The reason is the load sizing: 150 MΩ × 50 nA = 7.5 V of drop is impossible from a 2.5 V supply, so each branch is limited to ≈ VBAT/150 MΩ ≈ 17 nA (total ≈ 33 nA < 100 nA tail). The tail M2 is therefore pushed towards triode, net5 is close to ground, and M0 behaves as a common-source stage whose switching point is set by its own VGS at ≈ 14 nA (≈ 0.36 V), not by VREF.

Consequences: (1) the threshold inherits the VT temperature dependence of M0 (≈ −1 mV/°C at the gate, multiplied by 1/β ≈ 6.8 at the battery, i.e. several mV/°C, which over 5–45 °C could exceed the entire 200 mV window); (2) VREF/bandgap does not set the accuracy as intended; (3) the gm/Id targets (16 V^{-1} at 50 nA) are not actually realised in the pair. This is a hypothesis derived from hand analysis and the numbers above — it must be confirmed by reading net5, the M2 operating region and I(M0), I(M1) from the dcOp results, and by a 5–45 °C temperature sweep.

6.5 Current consumption (R.033)

The supply current was simulated with a DC sweep of VBAT (0–5 V) and the current through the VBAT source (V0) was plotted (Figure 7). The current is negative because it flows out of the source's positive terminal. Across the specified battery range the POR consumes:

- **549.9 nA at VBAT = 2.3 V** (cursor M1)
- **661.8 nA at VBAT = 3.2 V** (cursor M2)

The current grows almost linearly with VBAT (≈ 124 nA/V), dominated by the divider, and stays below the 1 µA allocation (R.033) over the whole 2.3–3.2 V range. The maximum, 0.66 µA, is ≈ 0.95 % of Iavg,max (69.4 µA). The curve reaches ≈ 1 µA at VBAT = 5 V, which is outside the operating range. The result is slightly lower than the hand estimate of 0.65–0.75 µA. Note that the current-starved operation described in Section 6.4 means the branch currents are lower than designed, so the current must be re-simulated after the loads are re-sized.

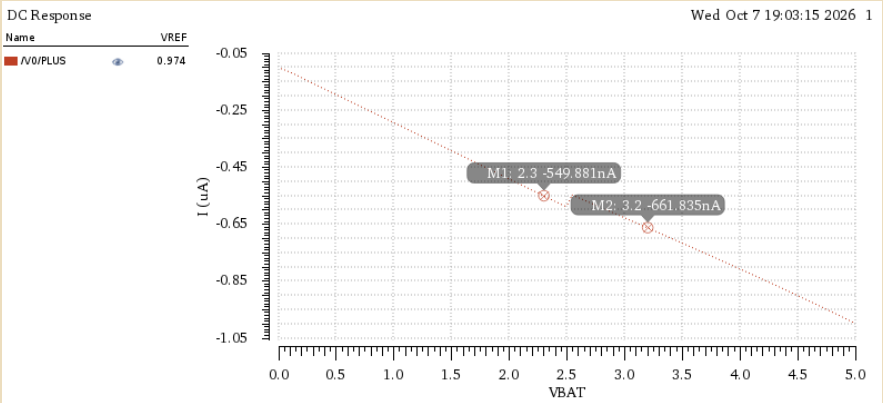


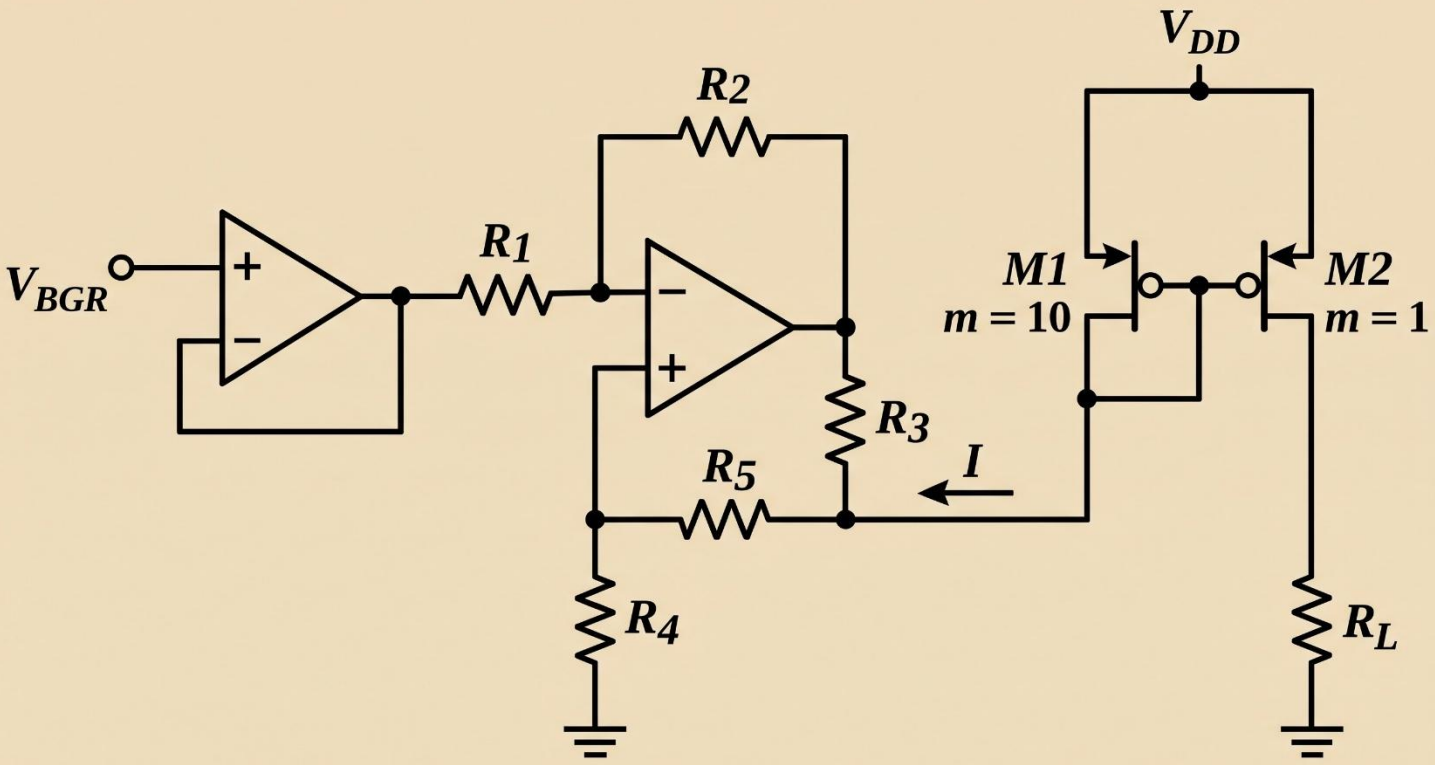
Figure 7 — POR supply current vs. VBAT (DC sweep); M1: 549.9 nA at 2.3 V, M2: 661.8 nA at 3.2 V

7. Next steps

The following actions are planned, in approximate priority order:

- Re-centre the thresholds: Vth_low (2.288 V in the latest transient) is below the 2.3 V limit and the 193 mV hysteresis leaves almost no slack. Reduce the hysteresis to ≈ 100–150 mV and move the nominal thresholds to ≈ 2.45 V / 2.33 V. Re-run the DC sweeps on the current netlist to compare them with the transient thresholds.
- Read dcOp (net5, M2 region, I(M0), I(M1), cmp) and run a 5–45 °C temperature sweep to confirm or reject the current-starved hypothesis; then re-size the loads ($R_{load} \approx \Delta V / I_{branch}$, e.g. 0.5 V / 50 nA = 10 MΩ instead of 150 MΩ), keeping the 16 V^{-1} / 50 nA pair and 100 nA tail.
- Decide the VREF source with the PMU team (most likely the bandgap, nominal ≈ 1.0 V) and re-derive the divider: β = VREF / Vth_rise (e.g. 1.0 / 2.45 = 0.408) with a hysteresis ratio around 1.05.
- Implement the explicit NMOS feedback switch M_FB driven by POR_INT (polarity of Section 3.1) and add an output buffer so POR is a full-swing digital signal; re-measure Vth_high / Vth_low / Vhyst and keep Icon < 1 µA (currently 0.55–0.66 µA).
- Verification: PVT (TT/FF/SS × 5–45 °C × VBAT 2.3–3.2 V) including ramp-rate sensitivity, and Monte-Carlo (≥ 200 runs) for Vth_high, Vth_low and Vhyst, compared with the 3σ ≈ 78 mV prediction of Section 4.5.
- Add trim bits TRIM<3:0> on the divider to pull the mean into the 2.3–2.5 V window if the σ or corner spread does not fit.
- Add the enable d2a_por_en_vbat with a power-down switch to reach R.004 (≤ 1 nA), or document why the monitor stays always-on and how the 1 nA budget is allocated at top level.
- Integrate POR with the bandgap, bias generator and LDO start-up sequence (total PMU start-up 5 ms), checking VBG_OK, glitch immunity for slow / noisy VBAT ramps and low-VBAT behaviour; add DFT hooks and top-level transients (Checkpoints 6–7).

Bias Generator



Topology stages

Buffer stage to avoid loading the BGR

Real V-to-I converter (loaded by PMOS diode connected)

Current Mirror (its output is robust 100nA)

Topology design

As we see in the last report, we need to design the amplifiers with high open loop gain. And also, we need to use low power consumption. Good efficiency and stability. We can design it by telescopic cascode OTA since we aren't interested in the CMIR or the output swing.

But the loading of the resistor will ruin the OTA DC biasing, so we will use class AB output stage, and we can bias it by translinear loop, "for more details about the translinear loop see the referred paper: Rail-to-Rail Input/Output Operational Amplifier for VLSI Cell Libraries by Hogervorst, Tero, Eschauzier, and Huisingh (IEEE JSSC, Dec. 1994)"

To bias the cascode transistors in the and then current source transistors in the circuit, we will implement a small circuit to bias them all, "for more details see how I implement it before by two ways in my referred report Op-amp report repo"

The current mirror will be designed to mirror $1\mu A$ to 100nA, and that's to reduce by factor of ten the errors from all of the V-to-I output current due to resistance mismatch, approximation error in the simplified current formula $I_{out} \approx -(R_2/R_1R_3)V_{in}$, static error in the buffer output. And also, the value of the current in μA allows to select the values of the resistors $R_1, R_2, R_3, R_4, and\ R_5$ in range of $1M\Omega$, and this is a good range since it's not very high to implement on the chip, and not very low comparable to the class AB output impedance.

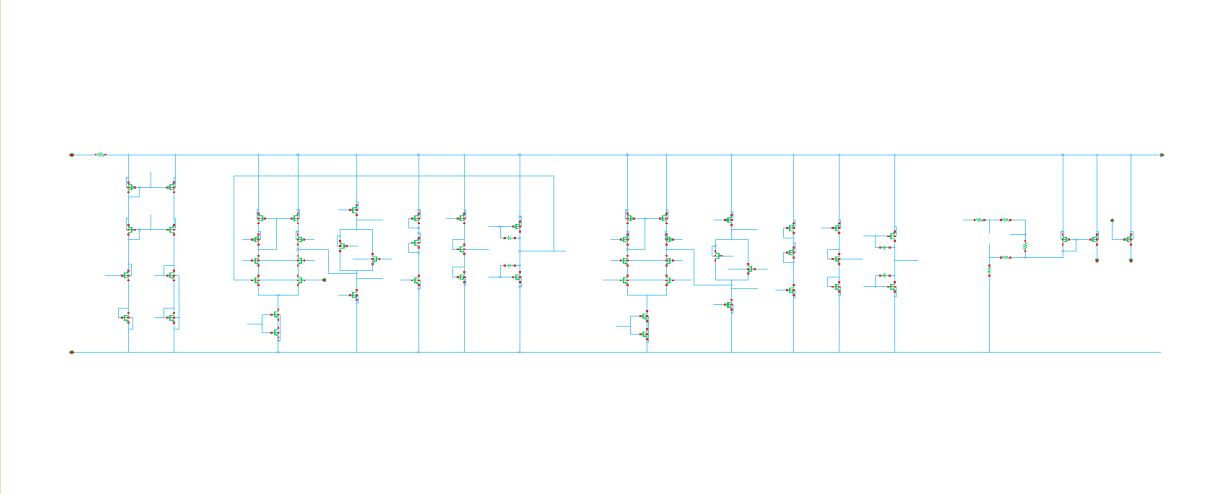
The current mirror implemented by simple current mirror to maximize the output headroom.

The values of the resistor will be selected as following:

$I_{out} = -\frac{R_2}{R_1 R_3} V_{in}, V_{in} = V_{BGR} = 1V \Rightarrow \frac{R_2}{R_1 R_3} = 1 \times 10^{-6}$, let $R_2 = 1M\Omega$, $R_1 = 1M\Omega \Rightarrow R_3 = 1M\Omega$

$R_1(R_3 + R_5) = R_2 R_4$, let $R_5 = 1M\Omega \Rightarrow R_4 = 2M\Omega$

$R_{L_{max}} = \frac{(R_4+R_3)}{R_3+R_4+R_5}\left(\frac{V_{sat}}{I_{outm}} - R_3\right)$, set $I_{outm} = 1\mu A, V_{sat} = \min(V_{DD}) = 2.3V \Rightarrow R_{L_{max}} \approx 0.975M\Omega$ which is very good to be loaded by PMOS diode connected



The designed schematic

If the telescopic cascode is PMOS input pair, let the input is $V_{BGR} = 1V \Rightarrow 1V = -V_{GS} + 4V_{DS}$ which can turn a cascode transistor off, but if it was NMOS input pair, $1V = V_{GS} + V_{DS-tail}$, $V_{DS-tail}$ can be very small if the tail current source biased by small voltage "small V^* ".

Also, if the telescopic cascode implemented by PMOS input pair, it will need PMOS head current source that will have different V_{GS} at each V_{DD} in the range of the battery, but if it implemented by NMOS input pair, the tail current source will have always constant V_{GS} as long as the BGR output voltage is constant.

The small bias circuit is on the left, then the buffer stage, amplifier stage, and in the right is the V-to-I converter and two output terminals.

Notice the resistor R_S on the left of the VDD pass, is used to shift the VDD range and to minimize the total power as we will see.

The bias circuit is composed of PMOS current source with width W_B and grounded gate, NMOS load and PMOS cascode current mirror with widths W_{BP}, W_{CP}, and NMOS cascode current mirror with widths W_{BN}, W_{CN}.

The cascode OTAs are biased by staked NMOS current source to reduce the power with area $W_T \times L$ and biased by $1V$ BGR.

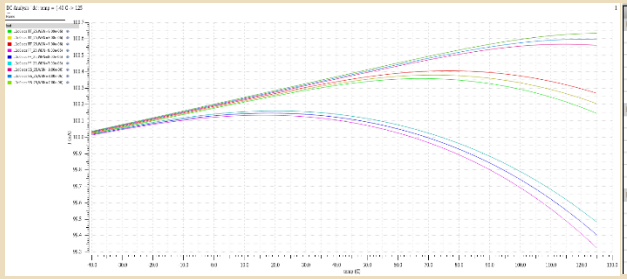
In the translinear loop and the class AB output stage, set the NMOS transistors with W_1 and the PMOS transistors with W_2, set the compensation capacitors to be 1pF, and then we may change them for stability.

1st set all transistors with max length ($L = 20\mu m$) to get the minimum current.

Note the PMOSs current mirror in the output is designed in the last report by ($L = 5\mu m, W = 2.6\mu m$).

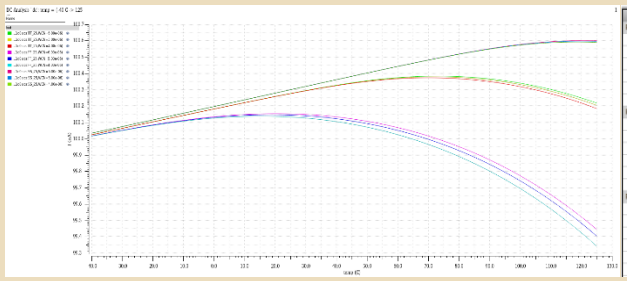
Set all of the $W_I, W_T, W_B, W_{1,2} = 0.4\mu m$ to get the minimum current, for the translinear loop and the class AB output stage we will need to drive a little more current for stability, so instead of increasing the width, we can reduce the length to optimize the area, for them set the length is $L/2$.

For $W_{BN}, W_{CN}, W_{BP}, W_{CP}$ we will sweep on them to select the values that gives lower variation in the output current with the lower power consumed across the corners and V_{bat} range.



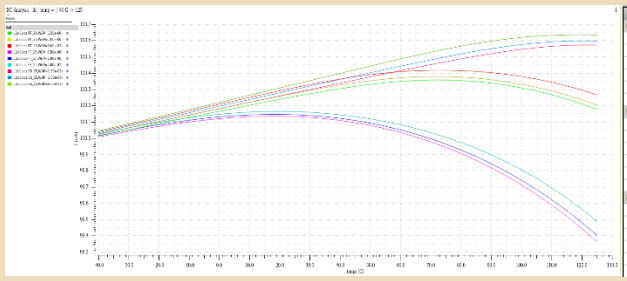
Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
Parameters: WBN=7u										
1	blocks:BG-tb:1	Iout								
1	blocks:BG-tb:1	I-variations				377.6p	678.1p	377.6p	678.1p	597.9p
1	blocks:BG-tb:1	Ipower								
1	blocks:BG-tb:1	max-I				2.135u	2.732u	2.423u	2.732u	2.135u
1	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
1	blocks:BG-tb:1	/VDPLUS								
Parameters: WBN=8u										
2	blocks:BG-tb:1	Iout								
2	blocks:BG-tb:1	I-variations				355.2p	742.1p	355.2p	742.1p	563.8p
2	blocks:BG-tb:1	Ipower								
2	blocks:BG-tb:1	max-I				2.134u	2.729u	2.421u	2.729u	2.134u
2	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
2	blocks:BG-tb:1	/VDPLUS								
Parameters: WBN=9u										
3	blocks:BG-tb:1	Iout								
3	blocks:BG-tb:1	I-variations				336.4p	808.5p	336.4p	808.5p	535.8p
3	blocks:BG-tb:1	Ipower								
3	blocks:BG-tb:1	max-I				2.134u	2.726u	2.419u	2.726u	2.134u
3	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
3	blocks:BG-tb:1	/VDPLUS								

W_{BN}



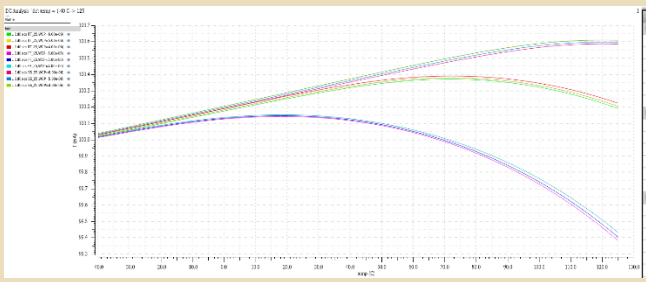
Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
Parameters: WCP=4u										
1	blocks:BG-tb:1	Iout								
1	blocks:BG-tb:1	I-variations				360.7p	719.2p	360.7p	719.2p	571.9p
1	blocks:BG-tb:1	Ipower								
1	blocks:BG-tb:1	max-I				2.116u	2.704u	2.399u	2.704u	2.116u
1	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
1	blocks:BG-tb:1	/VDPLUS								
Parameters: WCP=5u										
2	blocks:BG-tb:1	Iout								
2	blocks:BG-tb:1	I-variations				355.2p	742.1p	355.2p	742.1p	563.8p
2	blocks:BG-tb:1	Ipower								
2	blocks:BG-tb:1	max-I				2.134u	2.729u	2.421u	2.729u	2.134u
2	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
2	blocks:BG-tb:1	/VDPLUS								
Parameters: WCP=6u										
3	blocks:BG-tb:1	Iout								
3	blocks:BG-tb:1	I-variations				351.2p	758.4p	351.2p	758.4p	557.5p
3	blocks:BG-tb:1	Ipower								
3	blocks:BG-tb:1	max-I				2.149u	2.748u	2.438u	2.748u	2.149u
3	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
3	blocks:BG-tb:1	/VDPLUS								

W_{CN}



Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
Parameters: WBP=500n										
1	blocks:BG-tb:1	Iout								
1	blocks:BG-tb:1	I-variations				378.1p	674.8p	378.1p	674.8p	590.5p
1	blocks:BG-tb:1	Ipower								
1	blocks:BG-tb:1	max-I				2.053u	2.618u	2.321u	2.618u	2.053u
1	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
1	blocks:BG-tb:1	/VDPLUS								
Parameters: WBP=1u										
2	blocks:BG-tb:1	Iout								
2	blocks:BG-tb:1	I-variations				355.2p	742.1p	355.2p	742.1p	563.8p
2	blocks:BG-tb:1	Ipower								
2	blocks:BG-tb:1	max-I				2.134u	2.729u	2.421u	2.729u	2.134u
2	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
2	blocks:BG-tb:1	/VDPLUS								
Parameters: WBP=1.5u										
3	blocks:BG-tb:1	Iout								
3	blocks:BG-tb:1	I-variations				344.5p	769.7p	344.5p	769.7p	548.2p
3	blocks:BG-tb:1	Ipower								
3	blocks:BG-tb:1	max-I				2.179u	2.788u	2.475u	2.788u	2.179u
3	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
3	blocks:BG-tb:1	/VDPLUS								

W_{BP}

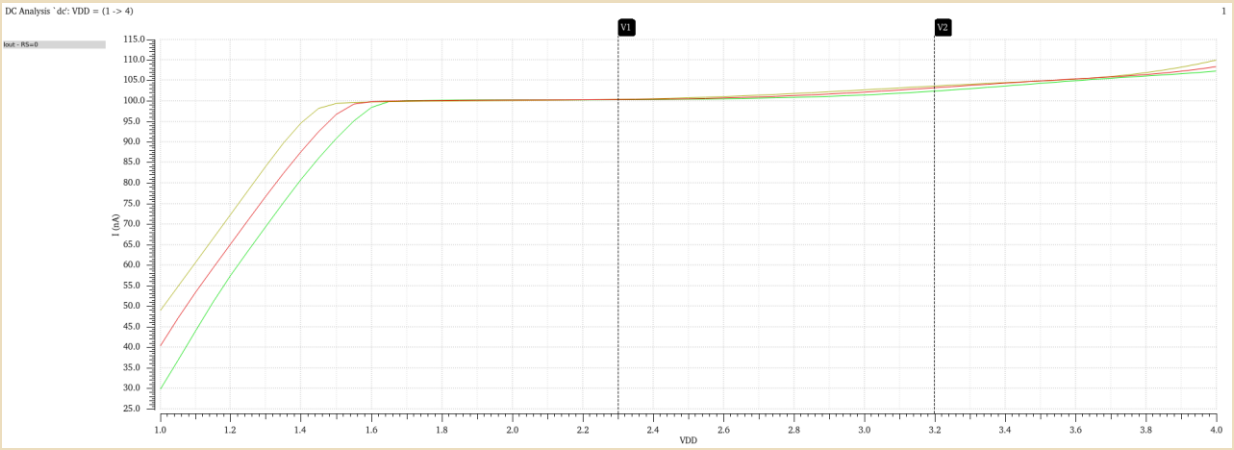


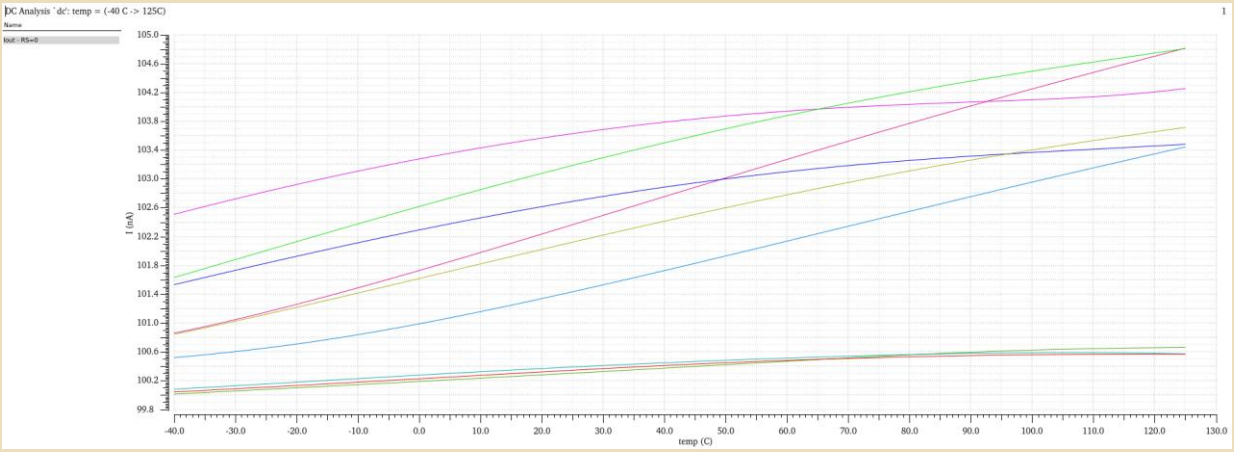
Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
Parameters: WCP=4u										
1	blocks:BG-tb:1	Iout								
1	blocks:BG-tb:1	I-variations				360.7p	719.2p	360.7p	719.2p	571.9p
1	blocks:BG-tb:1	Ipower								
1	blocks:BG-tb:1	max-I				2.116u	2.704u	2.399u	2.704u	2.116u
1	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
1	blocks:BG-tb:1	/VDPLUS								
Parameters: WCP=5u										
2	blocks:BG-tb:1	Iout								
2	blocks:BG-tb:1	I-variations				355.2p	742.1p	355.2p	742.1p	563.8p
2	blocks:BG-tb:1	Ipower								
2	blocks:BG-tb:1	max-I				2.134u	2.729u	2.421u	2.729u	2.134u
2	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
2	blocks:BG-tb:1	/VDPLUS								
Parameters: WCP=6u										
3	blocks:BG-tb:1	Iout								
3	blocks:BG-tb:1	I-variations				351.2p	758.4p	351.2p	758.4p	557.5p
3	blocks:BG-tb:1	Ipower								
3	blocks:BG-tb:1	max-I				2.149u	2.748u	2.438u	2.748u	2.149u
3	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
3	blocks:BG-tb:1	/VDPLUS								

W_{CP}

After some sweep on the widths, we select that, $W_{BN} = 8\mu m, W_{BP} = 1\mu m, W_{CN} = W_{CP} = 5\mu m$

To select the shifting resistance, R_S sweep on $V_{DD} \equiv V_{bat}$, set $R_S = 0$



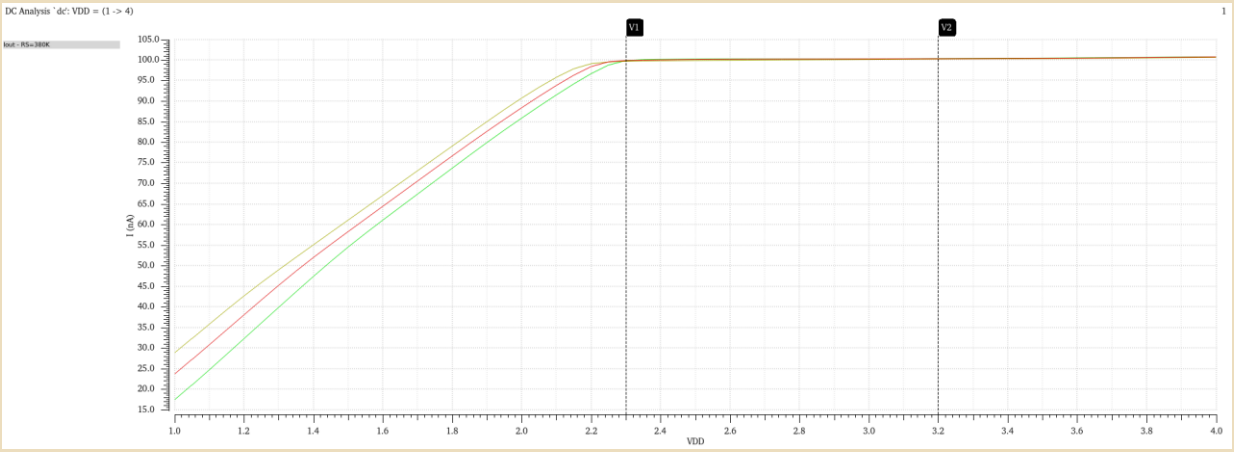


In our battery range ($2.3 \rightarrow 3.2V$), 1st the current is almost constant then start rising alittle. Notcie near to the edga at 1.6V the current almost constent with good window.

So, increasisng the shifting resistance will be helpful to reduce the current variations and also reduce the total power consumed.

As $R_S \uparrow$, the edge of the current is shifted to a higher value of VDD, the best value of R_S is that shift the edge of the current as near as to our window of work

Set $R_S = 380K\Omega$



Now our window of work is almost constant. Notcie, the edge of the current is shifted to start the constant range from value $\approx 2.3V$, which is better chooice for the shifting resistance in the prespictive of the power,

There is trad-off between the area efficiency, output current accuracy, and the total power consumed.

Last step of the design, We can see how the resutls changes with the length of the transistors L and minimize the area by selecting the minimum acceptable L that gives acceptable output current variations with acceptable total current consumed by the circuit.

Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
Parameters: VDD=2.3										
1	blocks:BG-tb:1	Iout								
1	blocks:BG-tb:1	I-variations				618.6p	1.075n	618.6p	1.075n	888.2p
1	blocks:BG-tb:1	Ipower								
1	blocks:BG-tb:1	max-I				1.536u	1.722u	1.608u	1.722u	1.536u
1	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
1	blocks:BG-tb:1	/V0/PLUS								
Parameters: VDD=3										
2	blocks:BG-tb:1	Iout								
2	blocks:BG-tb:1	I-variations				311.4p	587.3p	392.1p	311.4p	587.3p
2	blocks:BG-tb:1	Ipower								
2	blocks:BG-tb:1	max-I				1.77u	2.208u	1.977u	2.208u	1.77u
2	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
2	blocks:BG-tb:1	/V0/PLUS								
Parameters: VDD=3.2										
3	blocks:BG-tb:1	Iout								
3	blocks:BG-tb:1	I-variations				274.7p	672.4p	473.1p	274.7p	672.4p
3	blocks:BG-tb:1	Ipower								
3	blocks:BG-tb:1	max-I				1.945u	2.432u	2.177u	2.432u	1.945u
3	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
3	blocks:BG-tb:1	/V0/PLUS								

$L = 20\mu m$

After sweeping the value of L, the best chooice we finde is that $L = 8\mu m$, (set $L_B = 15\mu m$) to not affect the bias circuit.

Point	Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
Parameters: VDD=2.3										
1	blocks:BG-tb:1	Iout								
1	blocks:BG-tb:1	I-variations				171.4p	530.4p	171.4p	530.4p	381.6p
1	blocks:BG-tb:1	Ipower								
1	blocks:BG-tb:1	max-I				1.692u	1.963u	1.803u	1.963u	1.692u
1	blocks:BG-tb:1	VBGR								
1	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
1	blocks:BG-tb:1	/V0/PLUS								
Parameters: VDD=3										
2	blocks:BG-tb:1	Iout								
2	blocks:BG-tb:1	I-variations				356.3p	837.3p	668.7p	356.3p	837.3p
2	blocks:BG-tb:1	Ipower								
2	blocks:BG-tb:1	max-I				2.059u	2.646u	2.345u	2.646u	2.059u
2	blocks:BG-tb:1	VBGR								
2	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
2	blocks:BG-tb:1	/V0/PLUS								
Parameters: VDD=3.2										
3	blocks:BG-tb:1	Iout								
3	blocks:BG-tb:1	I-variations				607.3p	938.6p	862p	607.3p	938.6p
3	blocks:BG-tb:1	Ipower								
3	blocks:BG-tb:1	max-I				2.274u	2.908u	2.584u	2.908u	2.274u
3	blocks:BG-tb:1	VBGR								
3	blocks:BG-tb:1	VBGR-variations				8.25m	8.25m	8.25m	8.25m	8.25m
3	blocks:BG-tb:1	/V0/PLUS								

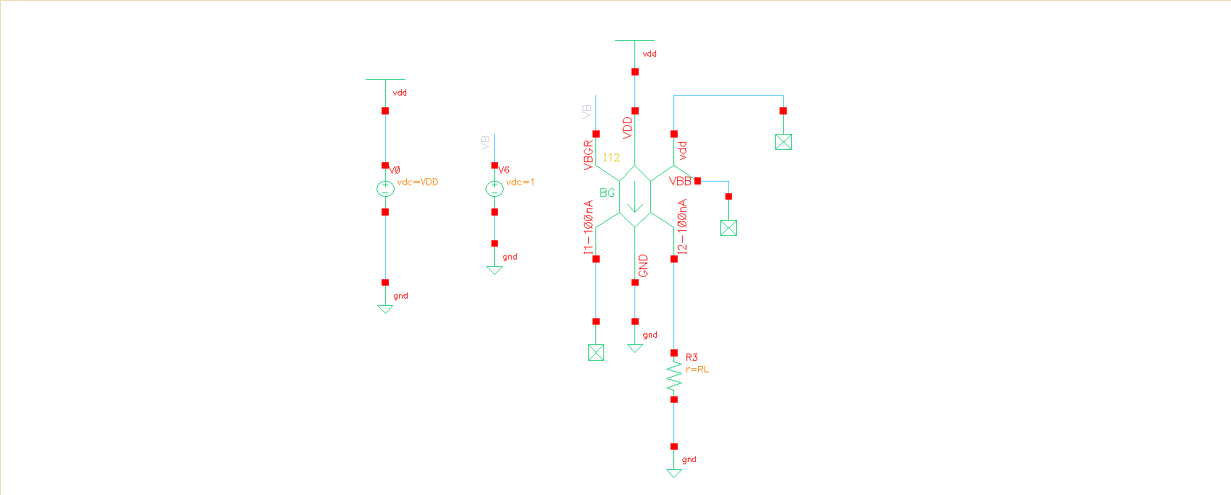
$L = 8\mu$

The reduction of the length significantly reduce the area, slightly increase both of the output current accuracy and the total power consumed which make it area efficient chooice and acceptable for the power prespective.

The total area used in the all above design is $1117.2\mu m^2$

Let out FOM $= \frac{1}{I \times Area} = \frac{1}{1.8\mu A \times 1117.2\mu m^2} = \frac{1}{1.8\times10^{-6} A \times 1117.2\mu m^2} \approx 497.27\ (1/A.\mu m^2)$

Test bench

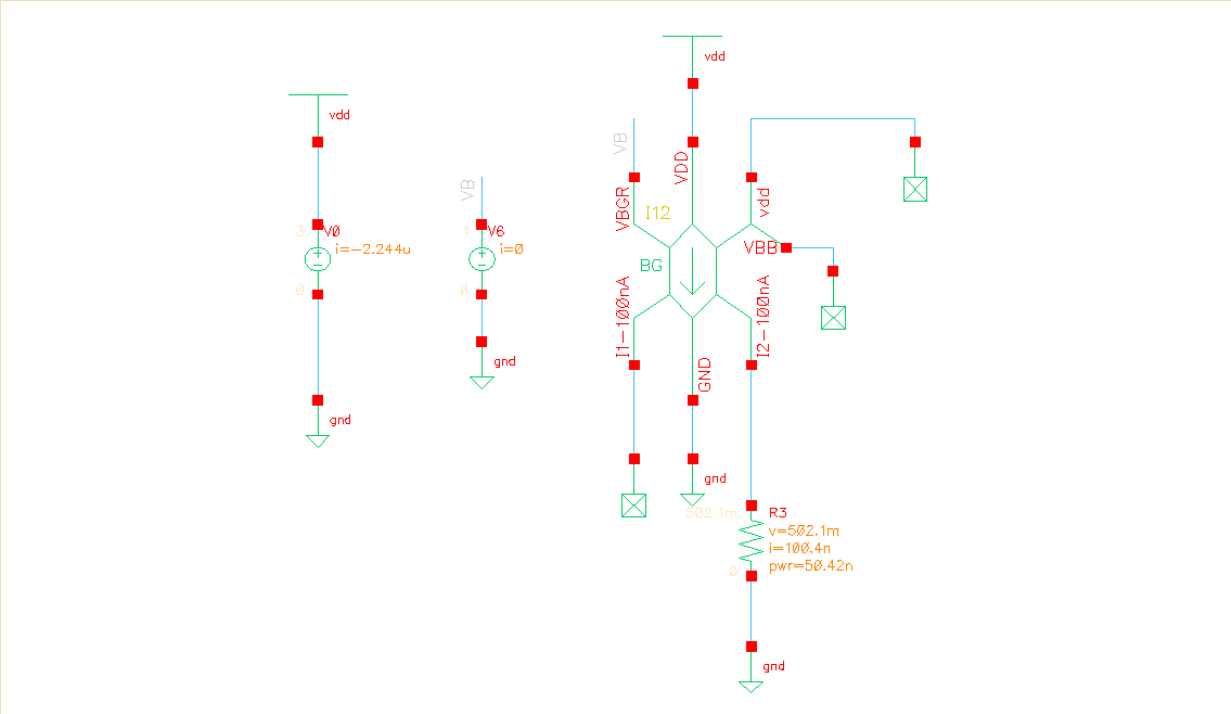


Test bench schematic

The power terminals VDD and GND, the input terminal VB that take 1V from the BGR. The output terminals I1-100nA and I2-100nA. The terminals vdd and VBB can be used to implement more branched if needed.

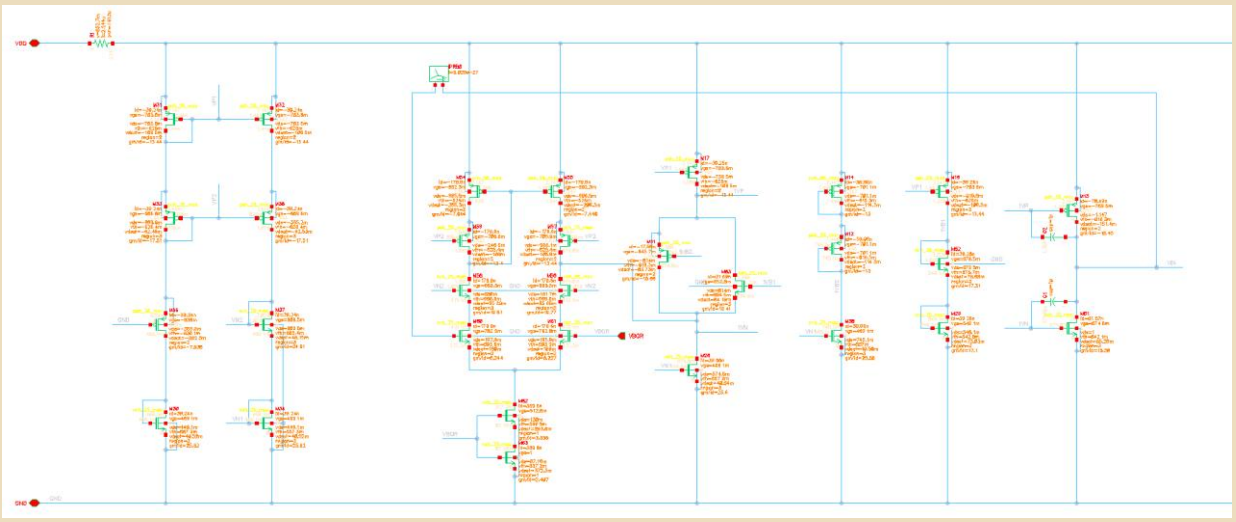
Best model for V_{BGR} is to set the temperature coefficient of the $V_B\{\alpha_1, \alpha_2 = 15 \times 10^{-6}, 1 \times 10^{-12}\}$ to vary with the temperature.

OP simulation

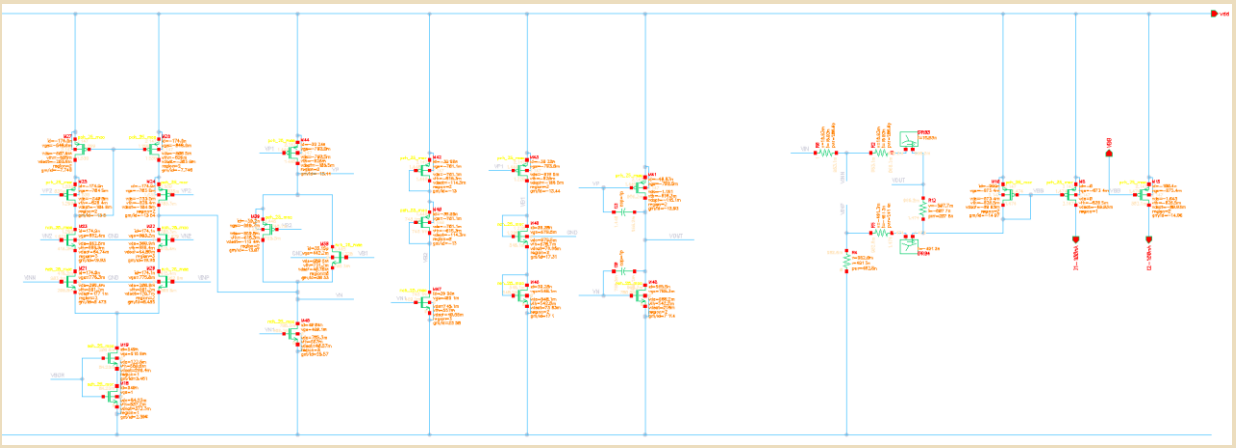


$R_L = 5M\Omega$

The output current is almost as desired, good power consumed



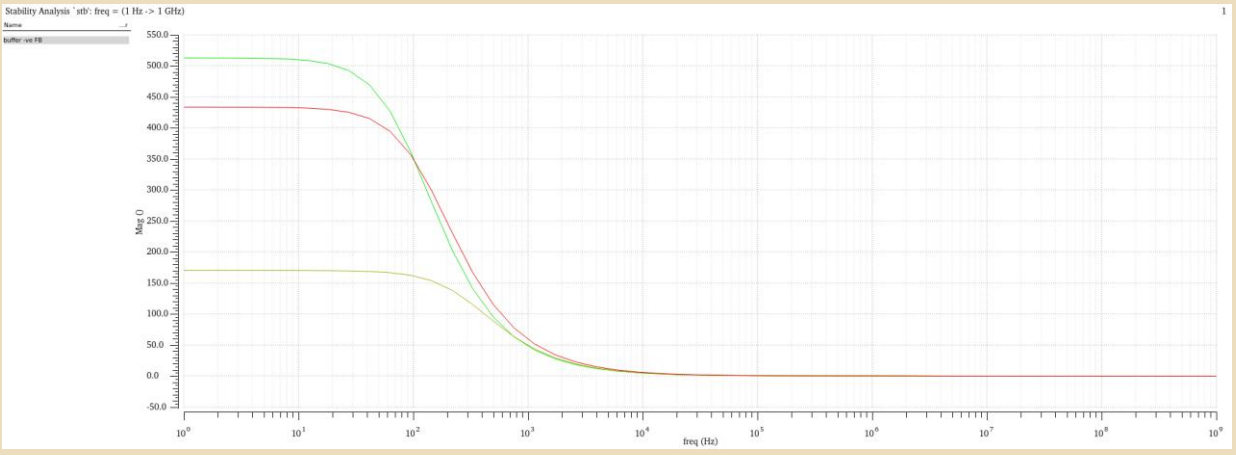
The bias circuit and the buffer OP simulation



The amplifier and the V-to-I converter OP simulation

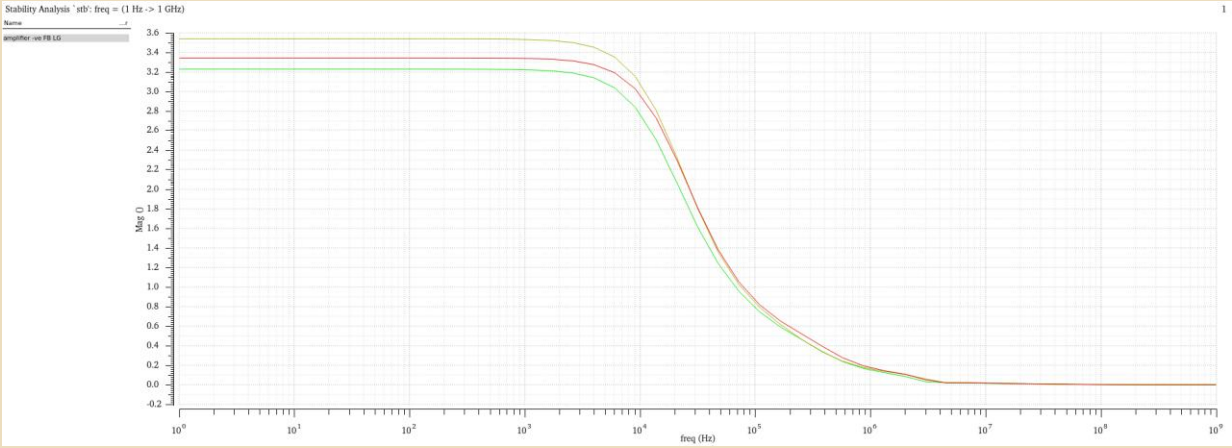
Stability simulation

Note, we have 3 feedback loops, one for the buffer -ve feedback, and two for the amplifier +ve and -ve feedback

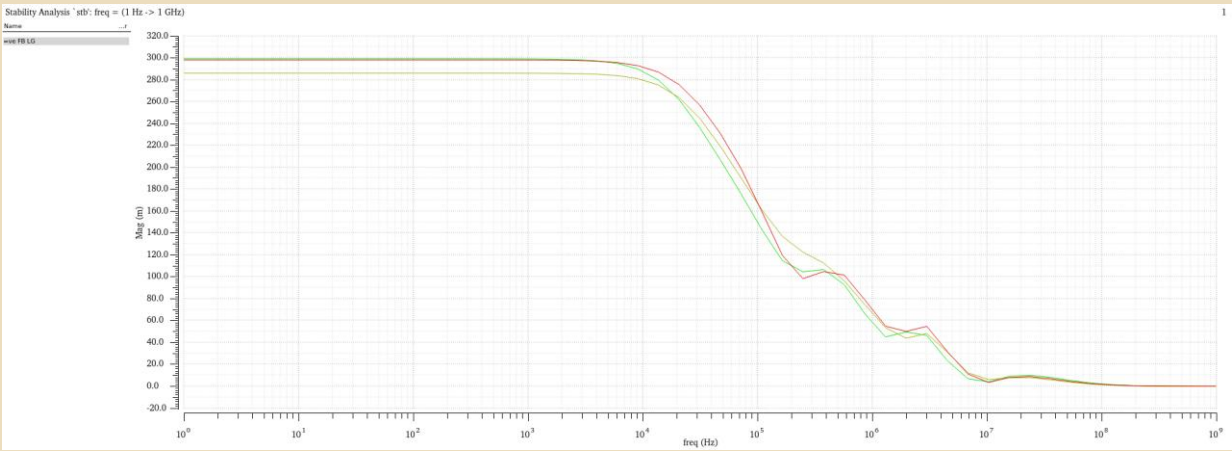


Buffer loop gain, gives high gain

For the amplifier loop gain, the -ve feedback loop gain has to be larger than the +ve feedback loop to make sure that the amplifier is stable.

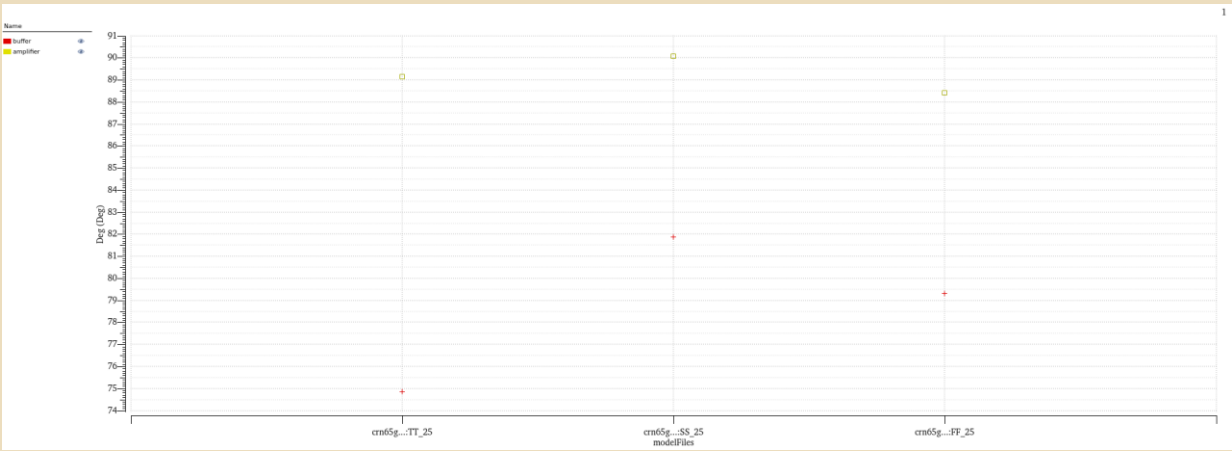


Amplifier -ve feedback loop gain



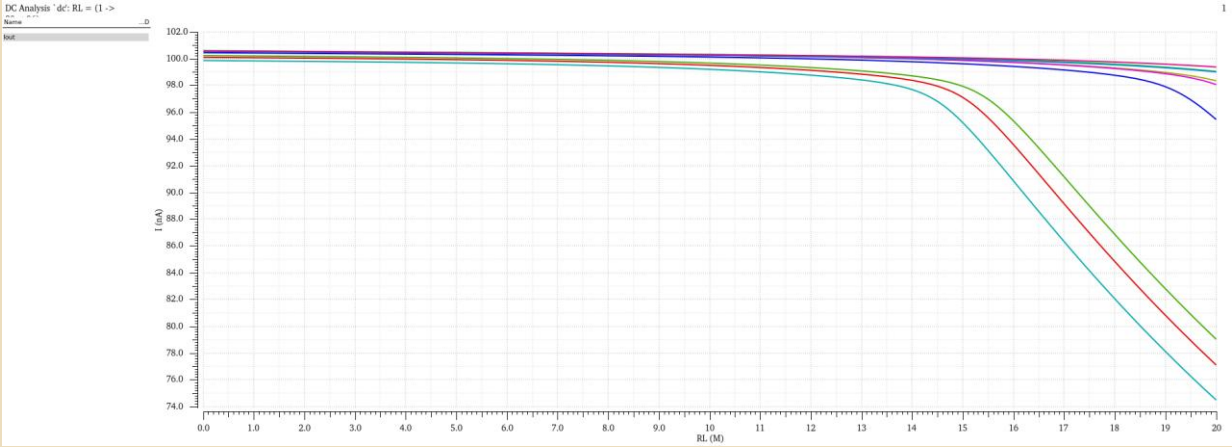
Amplifier +ve feedback loop gain

The loop gain of the -ve feedback is around 10 times of the +ve feedback loop gain.



Phase margin across the corners $C_C = 1pF$

They all have a good phase margin between 70° and 90°, so the value of the compensation capacitances 1pF is good for stability.



sweep on the load resistance R_L from $1\Omega \rightarrow 20M\Omega$

The above curve shows that, 1st the circuit provides robust current across the corners.

as R_L increases, V_{DS} of the PMOS mirror transistor decreases, at R_{Lmax} V_{DS} is smaller than the V^* and the transistor no longer in saturation.

We can get R_{Lmax} from the above graph at the edge of the current curve, we will find that at the $V_{bat} = 2.3V$ the maximum resistance that will not affect the PMOS operation region is around $12M\Omega$.

Notice as V_{bat} increases, the maximum load resistance also R_{Lmax} increases.

BGR

1. Summary

- **Goal:** all MOSFETs in subthreshold (weak inversion) except MR1, which must be in strong inversion (triode) [1].
- **Done:** extracted V_TH for the NMOS devices, the PMOS devices and MR1, and checked the operating region of every device. V_REF = 525.3 mV (nominal).
- **Status:** 7 of 13 devices are in the target region (M1, M2, M3, M5, M6, M7 and MR1).
- **Open:** M4 and the five PMOS mirror devices (ref1–ref5) are still in saturation (strong inversion). They need re-sizing.

2. Circuit

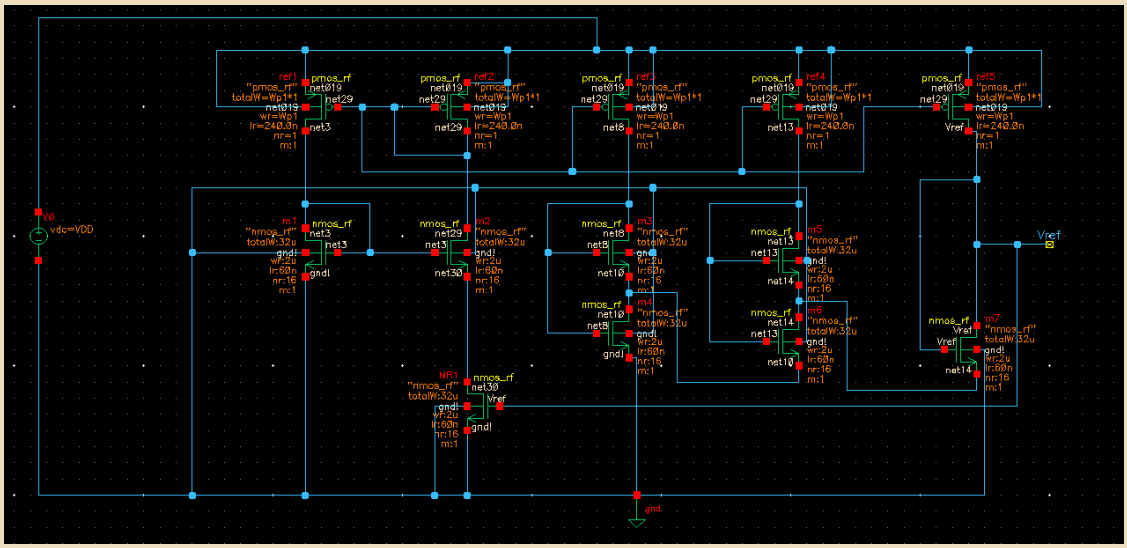


Fig. 1. Cadence schematic: NMOS M1–M7 and MR1, PMOS mirror ref1–ref5.

Current sizing (as labelled in the schematic): every NMOS has total W = 32 µm (2 µm × 16 fingers) and L = 60 nm. Every PMOS has W = Wp1 and L = 240 nm. Wp1 is the swept parameter in the plots (about 600–650 nm).

3. Key equations

Design variables are K = W/L and I_P, not V_{OV}, because M1–M7 operate with $V_{GS} < V_{TH}$ [1].

$$I_D = KI_0 e^{\frac{V_{GS}-V_{TH}}{\eta V_T}}\left(1 - e^{-V_{DS}/V_T}\right),\quad I_0 = \mu C_{ox}(\eta - 1){V_T}^2 \qquad (1)$$

Subthreshold current. Valid for $V_{DS} > 0.1$ V, where the last bracket is about 1.

$$V_{GS} = V_{TH} + \eta V_T ln\left(\frac{I_D}{KI_0}\right),\quad K = \frac{W}{L} \qquad (2)$$

Higher K at the same current gives lower V_{GS}. This is the main tuning knob.

$$V_{DSR1} = \eta V_T ln\left(\frac{K_2}{K_1}\right) \qquad (3)$$

V_{DS} across MR1 from the M1/M2 size ratio (K_2/K_1).

$$I_P = K_{R1}\mu C_{ox}(V_{REF} - V_{TH})\eta V_T ln\left(\frac{K_2}{K_1}\right) \qquad (4)$$

Bias current set by MR1 in deep triode.

$$V_{REF} = V_{TH} + \eta V_T ln\left(\frac{3I_P}{K_4 I_0}\right) + \eta V_T ln\left(\frac{2K_3 K_5}{K_6 K_7}\right) \quad (5)$$

Reference voltage.

$$V_{SG} = |V_{THP}| + \eta_p V_T ln\left(\frac{|I_D|}{K_p I_{0p}}\right), \; V_{OV,p} = V_{SG} - |V_{THP}| < 0 \quad (6)$$

PMOS mirror (subthreshold): same form, using V_{SG} and $|V_{THP}|$.

Region criteria: subthreshold requires $V_{OV} = V_{GS} - V_{TH} < 0$. MR1 requires $V_{OV} > 0$ (strong inversion) and $V_{DS} \ll V_{OV}$ (deep triode).

4. Cadence results

4.1 Threshold voltages

Device	V_{TH}	Note
NMOS m2	≈ 405 mV	read from the Fig. 2a marker
PMOS ref1	−319.992 mV	Fig. 2b
NMOS MR1	434.749 mV	Fig. 2c
All NMOS (M1–M7, MR1)	≈ 405–445 mV	spread in Fig. 2d
NMOS average	≈ 425 mV	estimate, please confirm

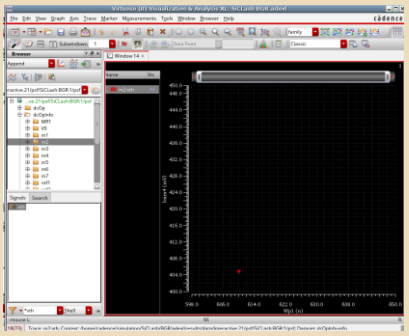


Fig. 2a. m2 V_{TH} vs Wp1

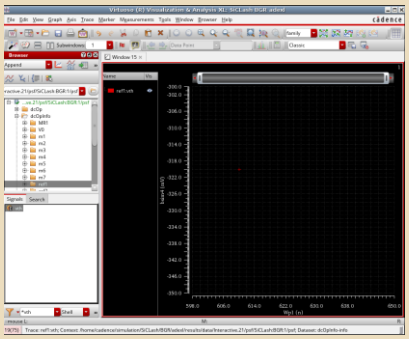


Fig. 2b. ref1 (PMOS) V_{TH} vs Wp1

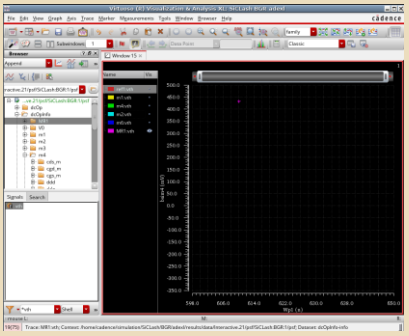


Fig. 2c. MR1 V_{TH} vs Wp1

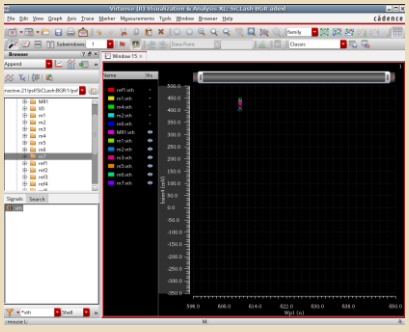


Fig. 2d. V_{TH} of all devices overlaid

4.2 Operating regions

Test	Output	Nominal	Spec	Weight	Pass/Fail
SiCLash:BGR:1	VDC("/Vref")	525.3m			
SiCLash:BGR:1	pv("m1" "region" ?result "dc...	3			
SiCLash:BGR:1	pv("m2" "region" ?result "dc...	3			
SiCLash:BGR:1	pv("m3" "region" ?result "dc...	3			
SiCLash:BGR:1	pv("m4" "region" ?result "dc...	2			
SiCLash:BGR:1	pv("m5" "region" ?result "dc...	3			
SiCLash:BGR:1	pv("m6" "region" ?result "dc...	3			
SiCLash:BGR:1	pv("m7" "region" ?result "dc...	3			
SiCLash:BGR:1	pv("ref1" "region" ?result "dc...	2			
SiCLash:BGR:1	pv("ref2" "region" ?result "dc...	2			
SiCLash:BGR:1	pv("ref3" "region" ?result "dc...	2			
SiCLash:BGR:1	pv("ref4" "region" ?result "dc...	2			
SiCLash:BGR:1	pv("ref5" "region" ?result "dc...	2			
SiCLash:BGR:1	pv("MR1" "region" ?result "dc...	1			

Fig. 3. ADE Assembler output: region of each device (BSIM4 code: 1 = triode, 2 = saturation, 3 = subthreshold).

Device	Code	Present region	Target	Status
M1, M2, M3, M5, M6, M7	3	Subthreshold	Subthreshold	OK
M4	2	Saturation	Subthreshold	Fix
ref1 – ref5 (PMOS)	2	Saturation	Subthreshold	Fix
MR1	1	Triode	Strong inversion, triode	OK

The region codes are the standard BSIM4 values. They are consistent with the circuit: MR1 reports triode and six NMOS report subthreshold.

5. Next steps

1. Move M4 into subthreshold. It carries $3 \cdot I_P$ at the same size as the other NMOS, so it has the highest current density. Increase its W/L, or lower I_P (Eq. 2).
2. Move ref1–ref5 into subthreshold. Their W/L is small (about 610 nm / 240 nm), so sweep Wp1 upward (up to 6 µm per device) and use multipliers for more width.
3. Check that MR1 is in deep triode: $V_{OV,MR1} = V_{REF} - V_{TH} \approx 525.3 - 434.7 \approx 91$ mV, so V_{DSR1} must be well below this value. Eq. 3 gives about 62 mV for $K_2/K_1 = 5$ and $\eta = 1.5$, which is not far below, so this needs checking.
4. Keep within the 65 nm limits (W max 6 µm, L max 240 nm). Large K values need parallel devices. K_{R1} from [1] (0.0267) is not realisable, so MR1 must be redesigned with the 65 nm model.
5. After each re-size, re-extract V_{TH}, the regions and V_{REF}, and record V_{GS}, V_{DS}, I_D and V_{TH} per device.

References

[1] K. Ueno, T. Hirose, T. Asai and Y. Amemiya, "A 300 nW, 15 ppm/°C, 20 ppm/V CMOS voltage reference circuit consisting of subthreshold MOSFETs," IEEE J. Solid-State Circuits, vol. 44, no. 7, pp. 2047–2054, Jul. 2009. https://eprints.lib.hokudai.ac.jp/dspace/bitstream/2115/39949/1/jssc$_{ueno2009}$.pdf

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